Evolution

Origin of life and Evolution of Life Forms - A Theory

1. The theory of spontaneous generation stated that
1) Life arose from living forms only
2) Life can arise from both living and non-living
3) Life can arise from non-living things only.
4) Life arises spontaneously, neither from living nor from the non-living.

2. Which of the following is an incorrect statement?
1) Life appeared 500 million years ago after the formation of earth.
2) According to the Big Bang theory, a thermonuclear explosion took place 10 billion years ago.
3) The theory of spontaneous generation stated that life came out of decaying and rotting matter like mud, straw, etc.
4) The theory of chemical origin regarding the origin of life is more or less accepted by most scientists.

3. According to Oparin and Haldane, the conditions on earth was
1) High temperature
2) Volcanic storms
3) Reducing atmosphere
4) All of these

4. Find the incorrect statement regarding the origin of life.
1) The first non-cellular forms of life originated 3 billion years ago.
2) Life originated 2 billion years after the origin of earth.
3) Pasteur, by careful experimentation, demonstrated that life comes only from pre-existing life.
4) The first cellular forms arose 2 billion years ago.

5. The chemical substance formed in S. L. Miller's experiment was
1) Nucleic acid
2) Amino acid
3) Trehalose
4) Palmitic acid

6. Mark the wrong statement.
1) The stellar distances are measured in light years.
2) There was reducing environment on primitive atmosphere.
3) Universe is around 20 billion years old.
4) Earth was formed 4.5 million years ago.

7. Which of the following statements is incorrect about the theory of special creation?
1) All living organisms that we see today were created as such.
2) Diversity of life was always the same since creation and will be the same in future.
3) Earth is 40,000 years old.
4) The theory was supported by Father Suarez.

8. The earliest organisms to appear on earth were
1) Aerobic chemoautotrophs
2) Anaerobic chemoheterotrophs
3) Anaerobic chemoautotrophs
4) Aerobic chemoheterotrophs

9. Evolutionary biology is

1) Origin of life from non-living matter.
2) Origin of life from pre-existing life.
3) Formation of diverse organic molecules from inorganic constituents.
4) Study of life forms on earth.

10. Find the correct match regarding Miller's experiment.
1) Temperature-80°C
2) Gases-H_2, NH_3, CH_4, O_2
3) Energy source-Electric discharge
4) Volume of glass chamber-50 L

What are the Evidences for Evolution? and What is Adaptive Radiation?

11. Paleontological evidences for evolution refer to the
1) Development of embryo
2) Homologous organs
3) Fossils
4) Analogous organs

12. Whales, bats, cheetah, and humans share similarities in the
1) Shape of jaw
2) Number of teeth
3) Pattern of bones of forelimbs
4) All of the above

13. Which of the following is incorrect about analogous organs?
1) They show superficial resemblance, though their internal structures are quite different.
2) They have similar function.
3) They are a result of convergent evolution.
4) They always develop in related organisms.

14. Phylogeny means
1) Developmental biology
2) Evolutionary history
3) Study of fossils
4) Molecular homology

15. Identify the wrong statement.
1) Heart and brain of vertebrates are homologous organs.
2) Homologous structures are a result of divergent evolution.
3) Tendrils of pea and grapevine are homologous structures.
4) All of the above.

16. Which of the following is not an example of analogous organs?
1) Sting of honeybee and scorpion
2) Flipper of penguins and dolphins.
3) Stem tendril of Passiflora and leaf tendril of Lathyrus aphaca
4) Thorn of Bougainvillea and tendril of Cucurbita

17. Read the given assertion and reason carefully and select the correct option.
Assertion: Whales, bats, cheetah, and human share similarities in the pattern of bones of forelimb.
Reason: All of them have humerus, radius, ulna, carpals, metacarpals, and phalanges.
1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.

3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

18. The pouched mammals of Australia survived because of
1) Lack of competition from any other mammal
2) Continental drift
3) Pouched mammals take better care of their young ones than placental mammals
4) Both (1) and (2)

19. Find the odd one with respect to their origin.
1) Spotted cuscus
2) Lemur
3) Numbat
4) Tasmanian tiger cat

20. Crocodile and penguin are similar to whale and dogfish in which one of the following features?
1) Possess bony skeleton
2) Lay eggs and guard them till they hatch
3) Have gill slits at some stage
4) Possess a single solid nerve cord

21. Fossils act as evidences of organic evolution. Which one of these features does not justify the feature of fossils?
1) Fossils present in the lower strata of earth are older than those present in the upper strata.
2) The study of fossils in different sedimentary layers indicates the geographical period in which they existed.
3) They are remains of decayed and decomposed body parts.
4) They are remains of hard parts of life forms that existed in the past.

22. The evolution of different species in a given area starting from a point and spreading to other geographical areas is known as
1) Migration
2) Adaptive radiation
3) Natural selection
4) Convergent evolution

23. Which of the following is an incorrect statement?
1) Organs that perform the same function, that are not similar in structural details and origin are homologous organs.
2) The branch of biology that deals with the study of fossils is palaeontology.
3) Darwin's finches are an example of adaptive radiation.
4) Adaptive radiation suggests divergent evolution.

24. Analogous organs arise due to
1) Divergent evolution
2) Artificial selection
3) Genetic drift
4) Convergent evolution

25. The study of fossils is useful evidence of evolution as it suggests
1) Life originated spontaneously from non-living matter.
2) Life came on earth as spores or Panspermia from other planets.
3) New forms of life arose at the same time in the history of earth.
4) Life forms varied over time and certain life forms are restricted to certain geological time spans.

26. Fossils are generally found in

1) Sedimentary rocks
2) Igneous rocks
3) Metamorphic rocks
4) Any type of rock

27. Read the given statements.
Statement I: Homologous organs have common ancestry and similar function.
Statement II: Analogous organs have different origin and dissimilar function.
1) Both Statement I and Statement II are true.
2) Both Statement I and Statement II are false.
3) Statement I is true, and Statement II is false.
4) Statement I is false, and Statement II is true.

28. Which of the following is a correct statement?
1) Homologous organs explain convergent evolution.
2) The modern-day amphibians that live on land and in water are supposed to have evolved from lobefins.
3) Sea weeds and few plants existed probably around 500 mya.
4) All of the above

29. Evolution of life shows that life forms had a trend of moving from
1) Land to water
2) Dryland to wet land
3) Fresh water to sea water
4) Water to land

Biological Evolution

30. Identify the wrong statement.
1) Lamarck's theory of evolution is called the theory of inheritance of acquired characters.
2) Darwin was influenced by the essays of Malthus.
3) According to Darwinism, evolution is a jerky and discontinuous process.
4) August Weismann gave the theory of "Continuity of Germplasm".

31. Which of the following does not favour Lamarckian concept of inheritance of acquired characters?
1) Limbless condition in snakes
2) Webbed toes in aquatic birds
3) Absence of eyes in cave salamander
4) Melanisation of peppered moth in industrial areas

32. Which of the following is incorrect regarding Darwinism?
1) Variations are small and directional.
2) Successful organisms leave more progeny than others.
3) All acquired characters are inherited to the next generation.
4) There is struggle for existence among animals for limited number of natural resources.

33. Darwin's theory of natural selection was based on certain observations. Which of the following is not an observation made by Darwin?
1) Natural resources are limited.
2) Populations are stable in size except for seasonal fluctuations.
3) Members of population are similar in all characteristics.
4) Most of the variations are inherited.

34. The phenomenon of industrial melanism supports

1) Mutation theory
2) Theory of abiogenesis
3) Natural selection
4) Allen's law

35. Animal husbandry and plant breeding programmes are the examples of
1) Reverse evolution
2) Artificial selection
3) Mutation
4) Natural selection

36. Match the options given in column I with those given in column II.

Column I		Column II	
A	Abiogenesis	(i)	Louis Pasteur
B	Reproductive fitness	(ii)	Hugo de Vries
C	Saltation	(iii)	Charles Darwin
D	Biogenesis	(iv)	Oparin-Haldane

1) A-(i), B-(ii), C-(iv), D-(iii)
2) A-(iv), B-(iii), C-(ii), D-(i)
3) A-(ii), B-(i), C-(iii), D-(iv)
4) A-(ii), B-(i), C-(iii), D-(iv)

37. Development and power of action of an organ are directly proportional to its use. The statement is correct according to
1) Darwinism
2) Lamarckism
3) Mendelism
4) Mutation theory

38. Which of the following is incorrect about Lamarckism?
1) Evolution of life forms occurred driven by the use and disuse of organs.
2) Giraffes elongated their neck to forage leaves on tall trees.
3) Giraffes with longer necks produced more offspring than those having shorter necks.
4) Giraffes passed on the acquired character of elongated neck to succeeding generations.

39. Which of the following is unrelated to Darwin's theory?
1) Survival of the fittest
2) Formation of new species due to large, discontinuous variations
3) Natural selection
4) Reproductive fitness

40. Identify the wrong statement.
1) Darwin was influenced by the work of Thomas Malthus on population.
2) Adaptive ability is inherited but has no genetic basis.
3) Some organisms are better adapted to survive in hostile environment.
4) Fitness is the result of the ability to adapt and get selected by nature.

Mechanism of Evolution and Hardy-Weinberg Equilibrium

41. Population is said to be non-evolving if all of the following features exist, except
1) Random mating
2) Large population size
3) Mutations are not taking place
4) New members migrate into this population from outside

42. Read the following statements and choose the incorrect one among them.

1) Disturbance in genetic equilibrium, or the Hardy Weinberg equilibrium, i.e., change in frequency of alleles in a population, would be interpreted as resulting in evolution.
2) When migration of a section of population to another place and population occurs, gene frequencies change in original as well as new population.
3) According to the Hardy-Weinberg principle, the allele frequencies in a population are unstable and vary from generation to generation.
4) Genetic recombination and natural selection can affect the Hardy-Weinberg equilibrium.

43. If nature selects individuals acquiring value other than the mean character value, the type of natural selection is called

1) Directional
2) Disruptive
3) Balancing
4) Stabilising

44. $(p + q)^2 = p^2 + 2pq + q^2 = 1$ represents an equation used in

1) Population genetics
2) Mendelian genetics
3) Biometrics
4) Molecular genetics

45. Match the column I with column II, and choose the correct option.

Column I		Column II	
a	Branching descent	1	Ichthyosaurs
b	Saltation	2	Tyrannosaurus
c	Fish like reptiles	3	Darwin
d	Biggest dinosaur	4	Hugo de Vries

1) a-3, b-4, c-1, d-2
2) a-3, b-4, c 2, d—1
3) a-4, b-3, c-1, d-2
4) a-4, b-3, c-2, d-1

46. The Hardy-Weinberg equation helps in knowing

1) Whether population size is constant
2) Whether evolution is occurring in a population
3) Behaviours of individuals in a population
4) Different characters of individuals in population

47. Identify the wrong statement with respect to the Hardy Weinberg principle.

1) The sum of all the allele frequencies is 1.
2) The gene pool remains constant, which is called genetic equilibrium.
3) There is no evolution if the allele frequencies are stable.
4) None of the above

48. Sometimes the change in allele frequency is so different in the new sample of population that they become a different species. The original drifted population becomes founder and the effect is called

1) Founder effect
2) Genetic effect
3) Mutual effect
4) Both (1) and (2)

49. Read the following statements and choose the incorrect option.

1) The frequency of AA individuals in a population is simply represented as 2p.

2) Allele frequencies in a population are stable and are constant from generation to generation.
3) Disturbance in genetic equilibrium, or Hardy Weinberg equilibrium, i.e., change of frequency of alleles in a population would be interpreted as resulting in evolution.
4) When frequency measured, differs from expected values, the difference (direction) indicates the extent of evolutionary change.

50. Hugo de Vries brought forth the idea of mutations -large differences arising suddenly in a population, based on his work on
1) Mint plant
2) Blue and green alga
3) Evening primrose
4) Cladophora

51. Evolution for Darwin was gradual while de Vries believed mutation caused speciation and hence called it
1) Elution
2) Recombination
3) Cloning
4) Saltation

52. Which of the following is incorrect regarding the work of Huge de Vries?
1) His work was based on evening primrose.
2) Mutations are responsible for evolution.
3) Mutations are small and directional.
4) Mutations caused speciation and hence, called saltation.

A Brief Account of Evolution and Origin and Evolution of Man

53. Consider the following statements and select the correct option stating which one is true (T) and which one is false (F).
(i) Tyrannosaurus was the biggest dinosaur about 20 feet in height and had huge fearsome dagger-like teeth.
(ii) Some land reptiles went back into water to evolve into fish-like reptiles probably 200 million years ago.
(iii) According to the theory of spontaneous generation, "panspermia" was transferred to different planets from outer space, including earth.
(iv) According to Darwin, there has been gradual evolution of life forms.
(v) Homology is based on convergent evolution, while analogy refers to a situation exactly opposite.

	(i)	**(ii)**	**(iii)**	**(iv)**	**(v)**
1)	T	T	T	F	F
2)	T	F	T	F	F
3)	T	T	F	T	F
4)	T	T	T	T	F

54. All of the following belong to the evolutionary line leading to evolution of mammals, except
1) Sauropsids
2) Synapsids
3) Pelycosaurs
4) Therapsids

55. Read the given statements of assertion and reason and select the correct option.
Assertion: Whales, dolphins, sea cows and seals are some mammals that live wholly in water.

Reason: Some of land reptiles went back into water to evolve into fish like reptiles probably 200 mya (e.g. Ichthyosaurs).

1) If both assertion and reason are true and the reason is the correct explanation of the assertion.
2) If both assertion and reason are true, but reason is not the correct explanation of the assertion.
3) If assertion is true, but reason is false.
4) If both assertion and reason are false.

56. Mark the wrong statement.

1) Giant ferns fell to form coal deposits.
2) In 1938, a fish caught in South Africa happened to be a coelacanth, which was thought to be extinct.
3) Amphibians lay thick-shelled eggs, which do not dry up in sun unlike those of reptiles.
4) By 500 million years ago, invertebrates were formed and active.

57. Bryophytes have evolved from

1) Psilophyton
2) Rhynia type
3) Chlorophyte ancestors
4) Tracheophyte ancestors

58. The most significant aspect of human evolution is

1) Erect posture
2) Opposable thumb
3) Increased brain size
4) Absence of tail

59. Read the following statements and identify the wrong one.

1) Homo habilis had a brain capacity of 650-800 cc.
2) Dryopithecus and Ramapithecus existed 50 million years ago.
3) The first human-like being or the earliest hominid was Homo habilis.
4) Prehistoric cave art developed around 18,000 years ago.

60. Find the mismatch among the given fossil men and their brain sizes.

1) Homo habilis—650-800 cc
2) Homo erectus-900 cc
3) Homo sapiens-900 cc
4) Australopithecus—500 cc

61. The Neanderthal man lived near

1) East and central Africa
2) East African grasslands
3) Ethiopia and Tanzania
4) East and central Asia

62. Match the columns A and B and select correct option.

Column I		Column II	
A	Ice age	(i)	Between 1,00,000-40,000
B	Origin of agriculture	(ii)	75,000-10,000
C	Neanderthal man	(iii)	18000
D	Pre-historic cave art	(iv)	10,000

1) (A)-ii, (B)–iv, (C)-i, (D)—iii
2) (A)-i, (B)—ii, (C)—iii, (D)-iv
3) (A)—ii, (B)—iii, (C)-i, (D)-iv
4) (A)—iii, (B)-iv, (C)-i, (D)–ii

63. Read the given statements and select the correct option.

Statement I: The skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull.

Statement II: Mammals are viviparous and protected their unborn young inside the mother's body.

1) Both Statement I and Statement II are true.
2) Both Statement I and Statement II are false.
3) Statement I is true, and Statement II is false.
4) Statement I is false, and Statement II is true.

64. Which of the following is an incorrect statement?

1) Agriculture came around 10,000 years back and human settlement started.
2) During the Ice Age between 75,000 and 10,000 years ago, modern Homo sapiens arose.
3) Homo sapiens arose in Africa and moved across continents and developed into distinct races.
4) Neanderthal man, with a brain size of 1400 cc, lived near east and central Asia between 40,000 and 10,000 years back.

Answer key

1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
3	2	4	2	2	4	3	2	4	3	3	3	4	2	3
16	**17**	**18**	**19**	**20**	**21**	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
4	1	4	2	3	3	2	1	4	4	1	2	2	4	3
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**	**41**	**42**	**43**	**44**	**45**
4	3	3	3	2	2	2	3	2	2	4	3	1	1	1
46	**47**	**48**	**49**	**50**	**51**	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
2	4	1	1	3	4	3	3	1	2	3	3	3	2	3
61	**62**	**63**	**64**											
4	1	1	4											